Algorithm Design - First Midterm
20-12-2024, *Time: 120 minutes*
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Instructions:

- You are given the question sheet and 4 sheets for your answers. The problems are 4.
- Before you start, write on the top of each sheet (1) your full name, (2) your student ID (*matricola*) and (3) the sheet number (1, 2, 3, and 4).
- If you use additional sheets as rough draft (*brutta*) for calculations, etc., you must hand it, along with the rest of the exam. You can keep the question sheet.
- Hand in the sheets in order **without attaching the sheets with each other.**
- When your pages are full, you can ask for extra pages; make sure that you also write there your name, student id, and the sheet number (5, 6, etc.). You must hand all the sheets that you receive.
- The rooms are small so it may be tempting to copy. Note that if we catch you copying in any way (copying from another student, using your phone, etc.) you will be automatically disqualified.

Problem 1

1. Two (directed or undirected) graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are called isomorphic, if there is a one-to-one function $f : V_1 \mapsto V_2$, such that for each $(u, v) \in E_1$ we have that $(f(u), f(v)) \in E_2$ and for each $(u, v) \in E_2$ we have that $(f^{-1}(u), f^{-1}(v)) \in E_1$. Informally, if G_1 and G_2 have the same number of nodes and the same connections between the corresponding nodes.

   Show that the *directed largest common subgraph* problem is NP-complete.

   Directed largest common subgraph problem:

   Input: Two directed graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, and a positive integer k.

   Question: Do there exist subsets $E_1' \subseteq E_1$ and $E_2' \subseteq E_2$ such that $|E_1'| = |E_2'| \geq k$ and such that the two subgraphs $G_1' = (V_1, E_1')$ and $G_2' = (V_2, E_2')$ are isomorphic?

2. Given an undirected graph $G = (V, E)$, a *dominating set* is a subset of nodes $S \subseteq V$ such that each node is either in S or is adjacent to at least one node in S.

   In the *dominating set problem* we are given an undirected graph G and a positive integer k. The question is whether G has a dominating set with size at most k. Show that the dominating set problem is NP-complete. (Hint: Give a reduction from vertex cover.)

Problem 2

In the *min-edge coloring* problem the goal is to color each edge of an undirected graph with as few colors as possible, such that no two edges that are adjacent to the same node have the same color. Give a 2-approximation algorithm for this problem.

Problem 3

The *max-3-hypergraph-cut problem* is defined as follows: the input instance is an unweighted 3-hypergraph $G = (V, E)$, where each hyperedge $e \in E$ is a set containing exactly three vertices (i.e., $e = \{u, v, z\}$ for some $u, v, z \in V$), and we ask to find a cut (V_1, V_2) such that the number of hyperedges having at least one node in V_1, and at least one node in V_2, is maximized. Provide a randomized approximation algorithm for this problem with an expected approximation ratio of at least 3/4.

Problem 4

The *weighted min-cost set cover problem* is defined on an universe U of n elements $U = \{e_i\}_{i=1}^n$ and a collection of m subsets of U, $S = \{S_j\}_{j=1}^m$, $S_j \subseteq U$. Each set $S_j \in S$ is also associated with a cost $c_j \geq 0$. Denote by $d_i = |\{S_j \in S : i \in S_j\}|$ the number of sets of S that contain element i, and let $d = \max_{i \in 1,\dots,n} d_i$.

1. Provide an LP formulation of the weighted min-cost set cover problem and its fractional (or linear) relaxation.
2. Design a polynomial time approximation algorithm based on deterministic rounding of the optimal fractional *LP* solution.
3. Show that the algorithm achieves an approximation ratio at most equal to d.